



RTL Design Sherpa

APB PIC 8259 Micro-Architecture Specification 1.0

January 4, 2026

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1 Pic 8259 Mas Index

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2 APB PIC 8259 - Overview

2.1 Introduction

The APB PIC 8259 is an 8259A-compatible Programmable Interrupt Controller with an APB interface. It provides interrupt management for legacy PC-compatible systems.

2.2 Key Features

Implemented in the current RTL:

- 8 interrupt inputs (`irq_in[7:0]`) with a single `int_out` line
- Fully-decoded 32-bit APB register file (no legacy A0 two-port model)
- Programmable priority with rotation (set-priority / rotate-on-EOI)
- Edge or level triggering
- Interrupt masking (IMR / OCW1)

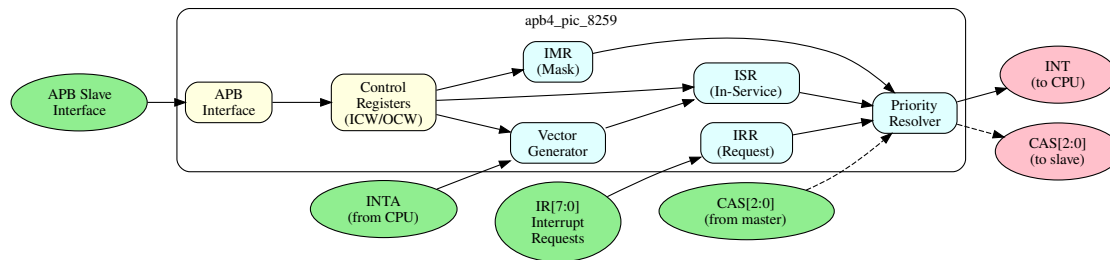
Register bits exist but are **not** functional in the current core (see the register map implementation notes): Master/Slave cascade (ICW3), Automatic EOI (ICW4 AEOI), polling mode (OCW3), special fully nested mode (ICW4 SFNM), and buffered mode (ICW4 BUF). There is also no INTA handshake or vector-output pin.

2.3 Applications

- PC-compatible interrupt management
- Legacy device support
- x86 system integration

2.4 Block Diagram

2.4.1 Figure 1.1: PIC 8259 Block Diagram

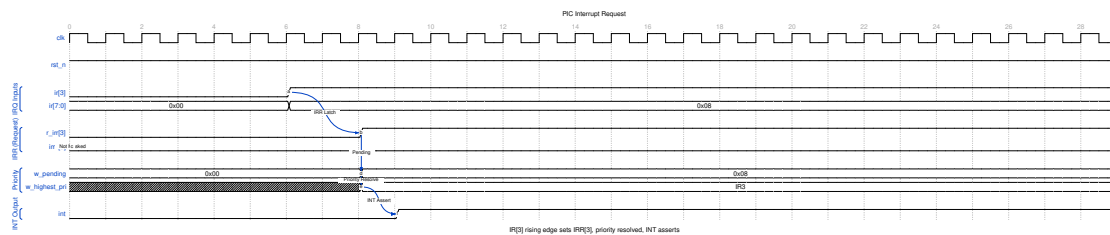


PIC 8259 Block Diagram

2.5 Timing Diagrams

2.5.1 Waveform 1.1: Interrupt Request

Shows an IRQ input assertion triggering the interrupt process.

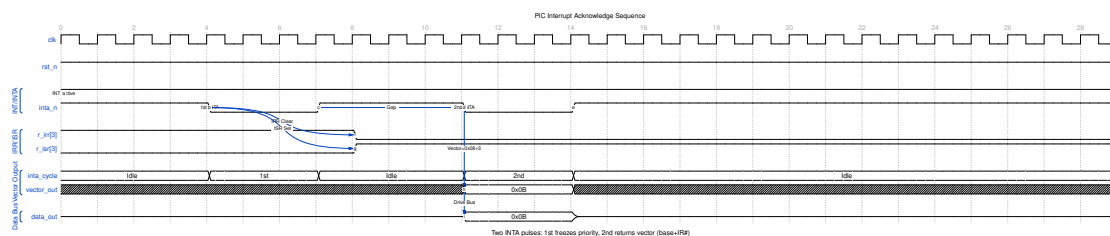


PIC Interrupt Request

When an IR pin asserts, the corresponding IRR bit is set. The priority resolver selects the highest priority unmasked interrupt and asserts INT to the CPU.

2.5.2 Waveform 1.2: Interrupt Acknowledge Sequence

The two-pulse INTA sequence from CPU to PIC.



PIC Interrupt Acknowledge

Note: the current RTL implements no INTA handshake. There are no `inta_n`, `cas`, or `sp_n/en_n` pins, `ISR` is never set, and the computed vector is not exposed to software. This waveform describes classic-8259A behavior that this block does not yet provide.

Software clears the in-service bit with an EOI command.

The diagram shows the following signals and their behavior over time:

- ck**: Clock signal, periodic square wave.
- rst_n**: Reset signal, active low, transitions from high to low and back to high.
- r_i**: Interrupt request signal, transitions from low to high.
- s_apb_psel**: APB select signal, transitions from `0x08` to `0x20`.
- s_apb_paddr**: APB address signal, transitions from `0x00000000` to `0x00000020`.
- s_apb_pwidth**: APB width signal, transitions from `0x00000000` to `0x00000020`.
- esi_cmd**: ESI command signal, transitions from `0x00000000` to `0x00000020`.
- specific_new**: Specific new signal, transitions from `0x00000000` to `0x00000020`.
- highest_isr**: Highest ISR signal, transitions from `0x00000000` to `0x00000020`.
- r_iwr**: Interrupt write signal, transitions from `0x00000000` to `0x00000020`.
- r_iwr[7:0]**: Interrupt write data signal, transitions from `0x00000000` to `0x00000020`.

Annotations in the diagram include:

- rst_n**: Reset
- r_i**: Interrupt request
- s_apb_psel**: APB select
- s_apb_paddr**: APB address
- s_apb_pwidth**: APB width
- esi_cmd**: ESI command
- specific_new**: Specific new
- highest_isr**: Highest ISR
- r_iwr**: Interrupt write
- r_iwr[7:0]**: Interrupt write data
- 0x08**: Value of `s_apb_psel`
- 0x00000000**: Value of `s_apb_paddr`
- 0x00000020**: Value of `s_apb_pwidth`
- 0x00000000**: Value of `esi_cmd`
- 0x00000020**: Value of `specific_new`
- 0x00000000**: Value of `highest_isr`
- 0x00000020**: Value of `r_iwr`
- 0x00000020**: Value of `r_iwr[7:0]`

Legend:

- ck**: Clock
- rst_n**: Reset
- r_i**: Interrupt request
- s_apb_psel**: APB select
- s_apb_paddr**: APB address
- s_apb_pwidth**: APB width
- esi_cmd**: ESI command
- specific_new**: Specific new
- highest_isr**: Highest ISR
- r_iwr**: Interrupt write
- r_iwr[7:0]**: Interrupt write data

Notes:

- 0x08: Value of `s_apb_psel`
- 0x00000000: Value of `s_apb_paddr`
- 0x00000020: Value of `s_apb_pwidth`
- 0x00000000: Value of `esi_cmd`
- 0x00000020: Value of `specific_new`
- 0x00000000: Value of `highest_isr`
- 0x00000020: Value of `r_iwr`
- 0x00000020: Value of `r_iwr[7:0]`

Timing diagram for PIC End-of-Interrupt (Non-Specific EOI). The diagram shows the sequence of events from a reset to the clearing of the highest priority ISR bit. The signals are: `ck` (Clock), `rst_n` (Reset), `r_i` (Interrupt request), `s_apb_psel` (APB select), `s_apb_paddr` (APB address), `s_apb_pwidth` (APB width), `esi_cmd` (ESI command), `specific_new` (Specific new), `highest_isr` (Highest ISR), `r_iwr` (Interrupt write), and `r_iwr[7:0]` (Interrupt write data). The diagram illustrates the sequence of events from a reset to the clearing of the highest priority ISR bit.

Non-specific EOI (0x20) clears the highest priority ISR bit. Specific EOI (0x60-0x67) clears a designated IR.

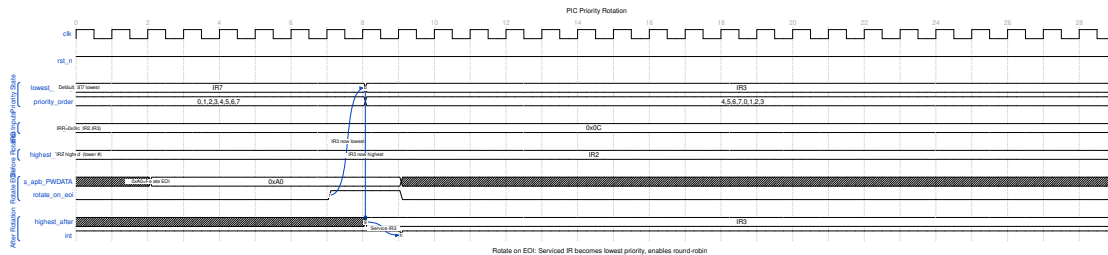
2.5.4 Waveform 1.4: Cascade Mode

Slave INT connects to master IR2. During INTA, master outputs cascade select (CAS) lines. Slave with matching ID provides the interrupt vector.

Note: cascade mode is not implemented in the current RTL. ICW3 is stored but inert, and there are no CAS or SP/EN pins. This waveform is illustrative of the classic-8259A architecture only.

2.5.5 Waveform 1.5: Priority Rotation

Automatic priority rotation for equal-service scheduling.



PIC Priority Rotation

Rotate-on-EOI (0xA0) makes the just-serviced IR the lowest priority, implementing round-robin scheduling among interrupt sources.

Note: the set-priority command (PIC_OCW2 = 0xC0-0xC7) does move the priority base in the current RTL, but rotate-on-EOI depends on the EOI path, which is inert because ISR is never set. See the register map implementation notes.

2.6 Register Summary

The block uses a fully-decoded 32-bit register file, not the legacy A0 two-port model. See [Chapter 5: Register Map](#) for full field definitions.

Offset	Register	Access	Description
0x00	PIC_CONFIG	RW	Global configuration (pic_enable, init_mode, auto_reset_init)
0x04	PIC_ICW1	WO	Initialization Command Word 1
0x08	PIC_ICW2	WO	Initialization Command Word 2 (vector base)
0x0C	PIC_ICW3	WO	Initialization Command Word 3 (cascade; inert)

Offset	Register	Access	Description
0x10	PIC_ICW4	WO	Initialization Command Word 4
0x14	PIC_OCW1	RW	Interrupt Mask Register (IMR)
0x18	PIC_OCW2	WO	EOI / priority command
0x1C	PIC_OCW3	WO	Special mask / read-select / poll
0x20	PIC_IRR	RO	Interrupt Request Register
0x24	PIC_ISR	RO	In-Service Register
0x28	PIC_STATUS	RO	Initialization state / diagnostics

The PIC is disabled at reset - firmware must set `pic_enable` (PIC_CONFIG bit 0) before any interrupt can be requested or delivered.

2.7 Interrupt Priority

IRQ	Default Priority
IR0	Highest (0)
IR1	1
IR2	2 (cascade input in a classic master; cascade not implemented here)
IR3	3
IR4	4
IR5	5
IR6	6
IR7	Lowest (7)

2.8 Priority Modes

- **Fixed Priority:** IR0 highest, IR7 lowest
- **Rotating Priority:** Lowest priority rotates after EOI
- **Specific Priority:** Programmable lowest priority

Next: 02_architecture.md (planned, not yet written)

3 APB PIC 8259 - Register Map

Unlike the original Intel 8259A, this block does **not** use the legacy two-port A0-based interface. It exposes a fully-decoded, 32-bit-aligned APB register file. Each ICW/OCW and every status register has its own dedicated offset; there is no A0 pin and no OCW3 read-select multiplexing. Only `regblk_addr[5:0]` is decoded, so the map repeats every 0x40 within the 4 KB APB window, and unmapped offsets (e.g. 0x2C) complete without PSLVERR.

3.1 Register Map

Offset	Register	Access	Reset	Description
0x00	PIC_CONFIG	RW	0x0000_0004	Global configuration and control (gates all operation)
0x04	PIC_ICW1	WO	—	Initialization Command Word 1 (edge/level, single, ICW4-needed)
0x08	PIC_ICW2	WO	—	Initialization Command Word 2 (interrupt vector base)
0x0C	PIC_ICW3	WO	—	Initialization Command Word 3 (cascade config; stored but inert)
0x10	PIC_ICW4	WO	—	Initialization Command Word 4 (mode bits)
0x14	PIC_OCW1	RW	0x0000_00FF	Operation Command Word 1 - Interrupt

Offset	Register	Access	Reset	Description
0x18	PIC_OCW2	WO	—	Mask Register (IMR) Operation Command Word 2 (EOI / priority command)
0x1C	PIC_OCW3	WO	—	Operation Command Word 3 (special mask, read-select, poll)
0x20	PIC_IRR	RO	0x0000_0000	Interrupt Request Register
0x24	PIC_ISR	RO	0x0000_0000	In-Service Register
0x28	PIC_STATUS	RO	—	Initialization state and diagnostics

Access notes: PIC_ICW1-ICW4, PIC_OCW2, and PIC_OCW3 are write-only in the RTL - their read-back paths are tied to zero, so reading these offsets returns 0x0000_0000. Only PIC_CONFIG, PIC_OCW1 (IMR), PIC_IRR, PIC_ISR, and PIC_STATUS return meaningful data on a read.

3.2 Global Configuration

3.2.1 PIC_CONFIG (Offset 0x00, RW)

The PIC is disabled out of reset. Firmware **must** set `pic_enable` before any interrupt request can propagate - while `pic_enable=0` the core holds IRR at 0 and forces `int_out` low.

Bits	Name	Access	Reset	Description
0	<code>pic_enable</code>	RW	0	Master enable (0=disabled, 1=enabled)
1	<code>init_mode</code>	RW	0	Initialization mode (0=operational, 1=init sequence)
2	<code>auto_reset_init</code>	RW	1	Automatically clear <code>init_mode</code> after ICW4 is written

Bits	Name	Access	Reset	Description
31:3	Reserved	RO	0	Reserved

3.3 Initialization Command Words (ICW)

All ICW registers are write-only; reading them returns 0.

3.3.1 PIC_ICW1 (Offset 0x04, WO)

Bit	Name	Reset	Description
0	IC4	0	1 = ICW4 needed, 0 = ICW4 not needed
1	SNGL	0	1 = single mode (no cascade), 0 = cascade mode
2	ADI	0	Call address interval (8080/8085 mode only)
3	LTIM	0	1 = level triggered, 0 = edge triggered
4	ICW1 Marker	1	Always 1 to identify ICW1 (8259A compatibility)
31:5	Reserved	0	Reserved (the RTL implements no bits above bit 4)

Note: the RTL stores only bits [4:0]. The legacy A7-A5 vector bits of an MCS-80 8259A are not implemented here.

3.3.2 PIC_ICW2 (Offset 0x08, WO)

Bits	Name	Reset	Description
7:0	vector_base	0x00	Interrupt vector base. The delivered vector is {vector_base[7:3], irq[2:0]}
31:8	Reserved	0	Reserved

3.3.3 PIC_ICW3 (Offset 0x0C, WO)

Cascade configuration. See the implementation note below - this register is stored but has no functional effect in the current RTL.

Master mode: | Bits | Description | |——|———| | 7:0 | Bitmap of IR lines that have a slave attached (1 = slave present) |

Slave mode: | Bits | Description | |——|———| | 2:0 | Slave ID (cascade input number 0-7) |

3.3.4 PIC_ICW4 (Offset 0x10, WO)

Bit	Name	Reset	Description
0	uPM	1	1 = 8086/8088 mode, 0 = 8080/8085 mode
1	AEOI	0	Automatic EOI mode (see implementation note)
3:2	BUF (M/S)	00	Buffered-mode select: 00=non-buffered, 10=buffered slave, 11=buffered master
4	SFNM	0	Special fully nested mode (see implementation note)
31:5	Reserved	0	Reserved

3.4 Operation Command Words (OCW)

3.4.1 PIC_OCW1 - Interrupt Mask Register (Offset 0x14, RW)

Bits	Name	Access	Reset	Description
7:0	imr	RW	0xFF	Interrupt mask for IRQ0-7: 1 = masked (disabled), 0 = unmasked (enabled)
31:8	Reserved	RO	0	Reserved

Reset masks all eight interrupts.

3.4.2 PIC_OCW2 (Offset 0x18, WO)

Bits	Name	Reset	Description
2:0	irq_level (L2-L0)	0	IRQ level for specific EOI or set-priority
4:3	Reserved	0	Reserved (OCW2 identifier field in a legacy 8259A)

Bits	Name	Reset	Description
7:5	eoi_cmd (R,SL,EOI)	0	EOI / rotation command (see table)
31:8	Reserved	0	Reserved

EOI / rotation command encodings (bits [7:5] = R, SL, EOI), as decoded by the RTL:

R	SL	EOI	Command
0	0	0	Rotate on auto EOI (clear)
0	0	1	Non-specific EOI
0	1	1	Specific EOI (clears the IR selected by L2-L0)
1	0	0	Rotate on auto EOI (set)
1	0	1	Rotate on non-specific EOI
1	1	0	Set priority (L2-L0 becomes lowest priority)
1	1	1	Rotate on specific EOI

3.4.3 PIC_OCW3 (Offset 0x1C, WO)

Bits	Name	Reset	Description
1:0	read_reg_cm d (RIS,RR)	00	Read-register select: 00=no action, 10=read IRR, 11=read ISR (see implementation note)
2	P	0	Poll command (see implementation note)
4:3	OCW3	01	Always 01 to identify OCW3

Bits	Name	Reset	Description
	Marker		
6:5	ESMM,SMM	00	Special mask mode: 10=reset special mask, 11=set special mask
31:7	Reserved	0	Reserved

3.5 Status / Readback Registers

3.5.1 PIC_IRR - Interrupt Request Register (Offset 0x20, RO)

Bits	Name	Reset	Description
7:0	irr	0	Pending interrupt requests for IRQ0-7 (1 = request pending)
31:8	Reserved	0	Reserved

IRR reflects requests before masking. In level mode the bits follow the IRQ pins; in edge mode they are set on a rising edge. See the implementation note below regarding clearing edge-triggered requests.

3.5.2 PIC_ISR - In-Service Register (Offset 0x24, RO)

Bits	Name	Reset	Description
7:0	isr	0	In-service bits for IRQ0-7 (1 = interrupt being serviced)
31:8	Reserved	0	Reserved

See the implementation note below - in the current RTL no path ever sets an ISR bit, so this register reads 0x00.

3.5.3 PIC_STATUS (Offset 0x28, RO)

Initialization-state and diagnostic readback. This register is the only way to observe the init sequence progress from software.

Bits	Name	Reset	Description
0	init_complete	-	1 = initialization complete, 0 = in init sequence
3:1	icw_step	-	Current ICW step (0 = not initialized, 4 = complete)

Bits	Name	Reset	Description
4	int_output	-	Current state of the INT output pin
7:5	highest_priority	-	Currently highest-priority IRQ (0-7)
31:8	Reserved	0	Reserved

3.6 Implementation Notes (RTL vs. classic 8259A)

The register file above matches the RTL exactly. Several classic-8259A behaviors implied by the register names are **not** implemented in the current core; they are documented here so firmware does not rely on them:

- **No INTA handshake / vector output.** The block has no `inta_n`, `cas[2:0]`, or `sp_n/en_n` pins - only `irq_in[7:0]` and `int_out`. The core computes an interrupt vector internally but leaves it on an unconnected wire reserved for future INTA support; no register exposes it to software.
- **ISR is never set (0x24 reads 0).** No acknowledge path sets an in-service bit, so `PIC_ISR` always reads 0x00. As a consequence, all `PIC_OCW2` EOI variants, ISR-based interrupt nesting/blocking, and special mask mode have no effect on live state.
- **Edge-triggered IRR has no clear-on-acknowledge path.** In edge mode an IRR bit, once set, is only cleared by reset, an `ICW1` write (re-initialization), or clearing `pic_enable`. Because EOI only touches the (always-zero) `ISR`, a pending edge interrupt can otherwise be removed only by masking it in the `IMR`. Level mode works normally because IRR follows the pin.
- **OCW3 read-select and poll are inert.** IRR and `ISR` are dedicated read-only registers at 0x20/0x24; the `OCW3 read_reg_cmd/poll` fields are decoded but not used, so the dedicated addresses make the read-select mechanism unnecessary.
- **Cascade, SFNM, buffered mode, and Auto EOI are stored but non-functional.** `ICW3` (cascade), `ICW4 SFNM/BUF`, and `ICW4 AEOI` are captured in registers but have no functional effect in the current core (`AEOI` performs no end-of-interrupt because `ISR` is never set).

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